> # CST1510 Programming for data communication and networks
>
> ## Week 5: File Handling and Data Structures including Pandas
>
> ## LAB ACTIVITY

ACTIVITY:

Threshold:

Activity 1: Writing Name and Date to File

Write a Python program that writes your name and today's date into a new file called `user_info.txt`.

Steps:

1. Open a new file called `user_info.txt` in write mode.
2. Write your name and the current date to the file.
3. Close the file after writing.

Activity 2: Appending an Address to File

Write a Python program that opens the `user_info.txt` file, appends an address to the end of the file, and then closes it.

Steps:

1. Open the `user_info.txt` file in append mode.
2. Write the address (e.g., "123 Main St, City, Country") to the end of the file.
3. Close the file after writing

Activity 3: Reading and Printing File Content

Write a Python program that opens the `Les_Brown.txt` file, reads the entire content, and prints it to the console.

Steps:

1. Open the `Les_Brown.txt` file in read mode.
2. Read the entire content of the file.
3. Print the content to the console.

Activity 4: Counting Lines in a File

Write a Python program that reads the `Les_Brown.txt` file and counts how many lines are in the file.

Steps:

1. Open the `Les_Brown.txt` file in read mode.
2. Read all lines into a list.
3. Use the `len()` function to count the number of lines.
4. Print the total number of lines.

Activity 5: Storing Clothing Data in a List

Write a Python program that reads data from the `clothing_stock.txt` file and stores each clothing item and its size in a list. Then print the list.

Steps:

1. Open the `clothing_stock.txt` file in read mode.
2. Read all lines and split them into clothing items and sizes.
3. Store the data in a list.
4. Print the list to the console.

Activity 6: Storing Unique Sizes in a Set

Write a Python program that reads the `clothing_stock.txt` file and stores all unique sizes in a set. Then print the set of sizes.

Steps:

1. Open the `clothing_stock.txt` file.
2. Extract only the size from each line and store it in a set.
3. Print the unique sizes.

Activity 7: Creating a Dictionary from Clothing Data

Write a Python program that reads the `clothing_stock.txt` file and creates a dictionary where the clothing item is the key and the count of each size is the value. Print the dictionary.

Steps:

1. Open and read the `clothing_stock.txt` file.
2. For each clothing item, count how many times each size appears.
3. Store the item and its count in a dictionary.
4. Print the dictionary.

Activity 8: Storing Favorite Movies

You are asked to create an aplication where users can store their favorite movies. Users can add, remove, or check if a movie is on their list of favorites. What data structure would you choose to store these favorite movies and why?

Example data:

```
["Inception", "The Matrix", "Interstellar", "The Dark Knight",
"Pulp Fiction"]


Steps:

1. Choose a data structure that allows adding, removing, and
checking movies easily.
2. Write a program that implements these operations.
3. Allow users to input and manipulate their movie list.
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```

Activity 9: Tracking Likes on Social Media Posts

You are building a tool that tracks the number of likes on social media posts. Users can like a post, remove their like, or check how many likes a post has. What data structure would you use to keep track of the likes on a post?

Example data:

```
{"John", "Emily", "Alex", "Sam"}
```

Steps:

1. Choose a data structure that can track unique likes efficiently.
2. Write a program that
   - allows users to like (i.e., add their name to the set of likes),
   - remove a like (i.e., remove their name from the set of likes),
   - display the total number of likes (i.e., the length of the set).
3. Implement functions to simulate these actions.

Typical:

Activity 10: Finding "Dream" Sentences in Les_Brown.txt

From the `Les_Brown.txt` file, read the content and identify all sentences that contain the word "dream". Write these sentences to a new file called `dream_sentences.txt`.

Steps:

1. Open and read the content of `Les_Brown.txt`.
2. Split the text into sentences.
3. Filter out sentences that contain the word "dream" (case-insensitive).
4. Write these sentences into the new file `dream_sentences.txt`.

Activity 11: Identifying and Converting Text

From the `Les_Brown.txt` file, read all content. Identify the part of the text that resonates with you, convert that part into uppercase, and write the modified text to a new file called `posiability.txt`.

Steps:

1. Open and read the contents of `Les_Brown.txt`.
2. Find the section of the text that resonates with you.
3. Convert the identified section to uppercase.
4. Replace the original section with the uppercase version in the text.
5. Write the updated text into a new file called `posiability.txt`.

Activity 12: Adding Line Numbers to Python Zen

Create a Python program that reads the content from the `python_zen.txt` file and adds line numbers to each line. Each line in the output file should start with a line number, followed by a dot and a space, and then the original line from the input file. Skip numbering the first line as it is the title. Save the result into a new file called `numbered_zen.txt`.

Steps:

1. Open the `python_zen.txt` file for reading.
2. Skip the first line (which is the title) when numbering.
3. For each subsequent line, prepend the line number followed by a dot and a space.
4. Write the modified content to a new file called `numbered_zen.txt`.

Activity 13: Reversing Clothing Stock List

Create a program that reads the `clothing_stock.txt` file and stores all the clothing items in a list. Then reverse the order of the items and write the reversed list into a new file called `reversed_clothing_stock.txt`.

Steps:

1. Read all clothing items into a list.
2. Reverse the list.
3. Write the reversed list into a new file.

Activity 14: Identifying Duplicate Entries in Clothing Stock

Create a program that reads the `clothing_stock.txt` file and identifies if there are any duplicate entries (both item and size). Use a set to find and print these duplicates.

Steps:

1. Read the file and store each item in a set.
2. Keep track of duplicate entries.
3. Print any duplicates found.

Activity 15: Counting Different Sizes per Clothing Item

Create a Python program that reads data from the `clothing_stock.txt` file and counts how many different sizes of each clothing item are available. Store this in a dictionary and print it.

Steps:

1. Read the file and extract both clothing items and sizes.
2. Count how many different sizes exist for each item.
3. Store the results in a dictionary and print it.

Activity 16: Managing Study Groups

You are responsible for managing a study group for a course. Each student can be part of multiple study groups, and you need to track which groups a student is enrolled in. Students can join or leave study groups. What data structure would you choose to manage this efficiently, and why?

Example data:

```
{"John": ["Math", "Physics"], "Emily": ["Physics", "Biology"],
"Alex": ["Math"]}

Steps:

1. Choose a data structure that allows quick access to the groups
each student belongs to.
2. Write a program that lets students join and leave study
groups.
3. Implement a function to display all groups a student is part
of.
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```

Activity 17: Managing and Shuffling Playlists

You are developing a music player that lets users create a playlist and shuffle it. Users should be able to add songs, remove songs, and shuffle the playlist randomly. What data structure would you use to represent the playlist, and why?

Example data:

```
["Song A", "Song B", "Song C", "Song D"]
```

Steps:

1. Choose a data structure that allows easy addition, removal, and shuffling of songs.
2. Write a program that simulates playlist operations (add, remove, shuffle).
3. Let users shuffle the playlist and display the new order.

Excellent:

Activity 18: Counting Words in Les_Brown.txt

Read the content from `Les_Brown.txt`, and then count the number of occurrences of each word in the text. Write the word count results to a new file called `word_count.txt`.

Steps:

1. Open and read the contents of `Les_Brown.txt`.
2. Split the content into words and count the frequency of each word.
3. Write the word count in the format: `word: count` to a new file called `word_count.txt`.

Activity 19 (List): Removing Duplicates from Clothing Stock

Write a Python program that reads the `clothing_stock.txt` file, removes any duplicate entries of clothing items with the same size, and then writes the unique entries to a new file called `unique_clothing_stock.txt`.

Steps:

1. Read the clothing items into a list.
2. Remove duplicates by storing them in a set.
3. Write the unique entries to a new file called `unique_clothing_stock.txt`.

Activity 20 (Set): Splitting Clothing Stock by Size

Create a Python program that reads the `clothing_stock.txt` file and splits the clothing items into separate sets based on their size (e.g., all XS items in one set, all M items in another set). Print each set.

Steps:

1. Read the file and separate items by their size.
2. Store each size in a different set.
3. Print each set of items.

Activity 21: Managing Event Tickets

You are developing a system to manage event tickets for concerts or sports events. Each event has a limited number of tickets available, and users can purchase tickets if they are still available. You need to track which users have purchased tickets for specific events and how many are still available. What data structure would you use for this task, and why?

Example data:

- Available tickets: `{"Concert": 100, "Football Game": 50, "Theater Show": 75}`
- Purchasers: `{"Concert": ["John", "Emily"], "Football Game": ["Alex"], "Theater Show": ["Sam", "John"]}`

Steps:

1. Choose a data structure to track ticket availability and purchasers efficiently.
2. Write a program that handles ticket purchases and updates the list of purchasers.
3. Implement functions to check availability and view who has purchased tickets.

Activity 22: Personal Finance App

You are creating a personal finance app that allows users to categorize their expenses and track spending by category (e.g., food, entertainment, travel). Users can add expenses, remove expenses, and see a summary of their spending. What data structure would you choose to store and organize the expenses, and why?

Example data:

```
{"Food": [25, 40, 15], "Entertainment": [50, 20], "Travel": [300, 150]}
```

Steps:

1. Choose a data structure that lets users easily add and track expenses by category.
2. Write a program that allows users to input expenses, remove them, and view their total spending by category.
3. Implement a function to display a summary of all spending.

Activity 23 (Dictionary and JSON): Grouping Clothing Stock by Size

Write a program that reads data from the `clothing_stock.txt` file and groups the clothing items by size using a dictionary (where the size is the key, and the value is a list of items). Write the dictionary into a JSON file.

Steps:

1. Read the file and extract sizes and items.
2. Group items by their size using a dictionary.
3. Write the dictionary into a JSON file.

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